

# Letter of Recommendation: Mubarak Ibrahim

**Date:** December 31, 2025

**To Whom It May Concern,**

I am writing to recommend **Mubarak Ibrahim [codeswot]** based on his contributions to the White Noise project, an open-source secure messaging application built on Nostr using the MLS protocol.

## Contribution Overview

**Project:** [White Noise](#) (Flutter/Dart + Rust)

**Period:** June 18, 2025 - December 31, 2025 (6.5 months)

**Impact:** 86 commits | 70+ pull requests | 25 issues created

## Assessment

Mubarak showed strong capabilities in frontend implementation and the ability to effectively integrate Rust-based backend systems into a Flutter app. His work on real-time messaging and notification systems shows he understands mobile application architecture and the challenges of building reliable communication systems.

His contributions were not limited to code. He actively identified issues, proposed solutions, and participated in the project's technical direction through well-documented issues and pull requests.

I recommend Mubarak for roles involving mobile development, real-time systems, or full-stack engineering positions where strong technical fundamentals and the ability to deliver complex features are valued.

## Technical Achievements

### Real-Time Systems & Backend Architecture

- Designed and implemented real-time chat message streaming using a custom provider architecture
- Built a notification service with deep-linking, allowing users to navigate directly into specific chats and invites from notifications
- Integrated Android WorkManager for background task execution, so messages are delivered even when the app is not in the foreground
- Developed a message polling system for both group chats and direct messages

### Core Feature Development

- Implemented legacy direct message support alongside MLS-based encrypted group chat creation
- Built message reply functionality with swipe gestures
- Created a bulk invite system with keypackage verification for secure group membership
- Developed leave group functionality with admin succession logic

## Performance Engineering

- Optimized group metadata loading and caching for faster UI rendering
- Improved notification polling intervals to balance responsiveness with battery efficiency
- Improved QR code scanning performance for contact sharing
- Implemented skeleton loaders to improve perceived application performance

## UI/UX Development

- Designed and built an app-wide toast notification manager for consistent user feedback
- Created a relay management interface with error handling
- Implemented chat list search with real-time filtering
- Built an emoji picker with responsive design using ScreenUtil for cross-device compatibility

## Infrastructure & Security

- Set up secure storage architecture for multi-account management
- Implemented camera permission handling using the permission\_handler package
- Built network relay error handling with clear user feedback

Sincerely,

Max Hillebrand

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*This recommendation is based on publicly available contribution data from the marmot-protocol/whitenoise GitHub repository.*